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Stress-reducing insole

Abstract

A stress-reducing insole includes a cushion unit, a frame and a superficial layer. The cushion unit includes air bags connected to one another. Each of the air bags includes a valve via which air is pumped into or released from the corresponding one of the air bags. The frame extends around the cushion unit and includes bores for receiving the valves. The superficial layer covers one of two opposite faces of the cushion unit.

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Background/Summary

BACKGROUND OF INVENTION
1. Field of Invention
(1) The present invention relates to an insole for footwear and, more particularly, to a stress-reducing insole.
2. Related Prior Art
(2) Painful areas of a sole of a foot are inevitably pressed when the foot is put in footwear. Thus, the pain in the painful areas of the sole is increased. Moreover, the recovery of the painful areas is compromised.
(3) As disclosed in U.S. Pat. Nos. 5,768,803, 4,793,078 and 5,438,768, inserts such as air bags and elastic blocks are inserted in insoles to provide some gaps between the insole and a user's sole. The gaps are intended to reduce pain in painful areas of the sole.
(4) U.S. Pat. No. 5,768,803 even discloses providing an insole with air bags. Some of the air bags corresponding to painful areas of a sole are deflated while the remaining ones of the air bags are inflated. The deflated air bags are intended to reduce stress in the painful areas of the sole.
(5) Disadvantageously, the inserts can easily be lost when they are detached from the insoles. Moreover, it has not been disclosed about how to inflate or deflate the air bags.
(6) The present invention is therefore intended to obviate or at least alleviate the problems encountered in the prior art.
SUMMARY OF INVENTION

(7) It is the primary objective of the present invention to provide effective and convenient stress-reducing insole.
(8) To achieve the foregoing objective, the stress-reducing insole includes a cushion unit, a frame and a superficial layer. The cushion unit includes air bags connected to one another. Each of the air bags includes a valve via which air is pumped into or released from the corresponding one of the air bags. The frame extends around the cushion unit and includes bores for receiving the valves. The superficial layer covers one of two opposite faces of the cushion unit.
(9) Other objectives, advantages and features of the present invention will be apparent from the following description referring to the attached drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) The present invention will be described via detailed illustration of the preferred embodiment referring to the drawings wherein:
(2) FIG. **1** is a perspective view of a stress-reducing insole according to the preferred embodiment of the present invention;
(3) FIG. **2** is another perspective view of the stress-reducing insole shown in FIG. **1**;
(4) FIG. **3** is an exploded view of the stress-reducing insole depicted in FIG. **1**;
(5) FIG. **4** is a cross-sectional view of the stress-reducing insole taken along a line IV-IV shown in FIG. **2**;
(6) FIG. **5** is a cross-sectional view of the stress-reducing insole taken along a line V-V shown in FIG. **2**;
(7) FIG. **6** is a cross-sectional view of the stress-reducing insole shown in FIG. **4** while inflated;
(8) FIG. **7** is a cross-sectional view of the stress-reducing insole shown in FIG. **4** while deflated;
(9) FIG. **8** is a cross-sectional view of the stress-reducing insole shown in FIG. **5** while supporting a user's foot;
(10) FIG. **9** is a cross-sectional view of the stress-reducing insole and the foot in another shown in FIG. **1**;
(11) FIG. **10** is a perspective view of a stress-reducing insole according to the second embodiment of the present invention;
(12) FIG. **11** is a perspective view of a stress-reducing insole according to the third embodiment of the present invention; and
(13) FIG. **12** is an exploded view of the stress-reducing insole depicted in FIG. **11**.
DETAILED DESCRIPTION OF EMBODIMENTS
(14) Referring to FIGS. **1** through **5**, a stress-reducing insole includes a cushion unit **10**, a frame and a superficial layer **50** according to a first embodiment of the present invention. Preferably, the frame consists of two semi-frames **20** and **30**.
(15) The cushion unit **10** includes air bags **12** independent of one another. The air bags **12** are connected to one another by connectors **13**. Each of the air bags **12** includes a valve **14**. The valves **14** are configured corresponding to a ball needle **15** that is often used with an air pump to pump a ball for example.
(16) The semi-frame **20** is made of an elastic, flexible and soft material suitable for insoles. The semi-frame **20** includes an internal face **22** shaped in compliance with a semi-periphery of the cushion unit **10**. Bores **24** are made in the internal face **22** of the semi-frame **20**. The semi-frame **20** includes two connective portions **23** at two ends.
(17) The semi-frame **30** is made of an elastic, flexible and soft material suitable for insoles. The semi-frame **30** includes an internal face **32** shaped in compliance with another semi-periphery of the cushion unit **10**. Bores **35** are made in the internal face **32** of the semi-frame **30**. The semi-

frame **30** includes two connective portions **34** at two ends.
(18) The connective portions **23** of the semi-frame **20** are connected to the connective portions **34** of the semi-frame **30** so that the semi-frames **20** and **30** are joined to provide the frame around the cushion unit **10**. The connective portions **23** of the semi-frame **20** are overlapped with the connective portions **34** of the semi-frame **30**. The bores **24** and **35** receive the valves **14**.
(19) The superficial layer **50** is made of an elastic, flexible and soft material suitable for insoles. The superficial layer **50** is coated on the cushion unit **10** and the semi-frames **20** and **30**. The superficial layer **50** is attached to the cushion unit **10** and the semi-frames **20** and **30** by adhesive for example.
(20) Referring to FIG. **6**, the ball needle **15** is inserted in a selected one of the air bags **12** via the corresponding valve **14**. Air is pumped into the selected air bag **12** via the ball needle **15**.
(21) Referring to FIG. **7**, the ball needle **15** is inserted in a selected one of the air bags **12** via the corresponding valve **14**. Air is released from the selected air bag **12** via the ball needle **15**.
(22) Referring to FIGS. **8** and **9**, some of the air bags **12** aligned to the painful areas **60** of a user's sole are deflated. Thus, stress in the painful areas **60** of a sole is reduced.
(23) Referring to FIG. **8**, the superficial layer **50** is used as a lower layer in contact with an outsole **51**. Thus, the user's sole is in direct contact with the cushion unit **10**. The inflated air bags **12** support the user's sole while the deflated air bags **121** are kept from the painful areas **60** of the user's sole, thereby reducing the stress in the painful areas **60** of the user's sole.
(24) Referring to FIG. **9**, the superficial layer **50** is used as an upper layer. The user's sole is in contact with the superficial layer **50**. The inflated air bags **12** support the user's sole while the deflated air bags **121** are kept from the painful areas **60** of the user's sole, thereby reducing the stress in the painful areas **60** of the user's sole.
(25) How much each of the air bags **12** is inflated or not at all is determined by how much the stress in a corresponding one of the painful areas **60** of the user's sole must be reduced and how much support the user's sole needs.
(26) Referring to FIG. **10**, there is shown a stress-reducing insole according to a second embodiment of the present invention. The second embodiment is identical to the first embodiment except for that the cushion unit **10** includes air bags **12** in different shapes in a different arrangement.
(27) Referring to FIGS. **11** and **12**, there is shown a stress-reducing insole according to a third embodiment of the present invention. The third embodiment is identical to the first embodiment except for including a continuous frame **70** instead of the frame that consists of the semi-frames **20** and **30**. The frame **70** includes bores **71** for receiving the valves **14**.
(28) The present invention has been described via illustration of the embodiments. Those skilled in the art can derive variations from the embodiments without departing from the scope of the present invention. Therefore, the embodiments shall not limit the scope of the present invention defined in the claims.

Claims

1. A stress-reducing insole comprising: a cushion unit (**10**) having an outer perimeter, the cushion unit comprising air bags (**12**), wherein each of the air bags (**12**) comprises a valve (**14**) via which air is pumped into or released from the corresponding one of the air bags (**12**); a frame extending around the outer perimeter of the cushion unit (**10**) such that the cushion unit substantially fills an area encompassed by the frame and comprising two semi-frames (**20**, **30**), wherein each of the semi-frames (**20**, **30**) comprises bores (**24**, **35**) for receiving the valves (**14**); and a superficial layer (**50**) coated on the frame.
2. A stress-reducing insole comprising: a cushion unit (**10**) comprising independent air bags (**12**) and connectors (**13**) for connecting the air bags (**12**) to one another, wherein each of the air bags

(**12**) comprises a valve (**14**) via which air is pumped into or released from the corresponding one of the air bags (**12**); and two semi-frames (**20**, **30**) each of which comprises: an internal face (**22**, **32**) shaped in compliance with a semiperiphery of the cushion unit (**10**); bores (**24**, **35**) for receiving the valves (**14**); and two connective portions (**23**, **34**) formed at two ends, wherein the connective portions (**23**) of one of the semi-frames (**20**) are connected to the connective portions (**34**) of the remaining one of the semi-frames (**30**) so that the semi-frames (**20**, **30**) are joined to provide a frame around the cushion unit (**10**).

3. The stress-reducing insole according to claim 2, further comprising a superficial layer (**50**) for covering one of two opposite faces of the frame.

4. The stress-reducing insole according to claim 2, wherein the connective portions (**23**) of one of the semi-frames (**20**) are overlapped with the connective portions (**34**) of the remaining one of the semi-frames (**30**).

5. A stress-reducing insole comprising: a cushion unit (**10**) having an outer perimeter, the cushion unit comprising independent air bags (**12**) connected to one another, wherein each of the air bags (**12**) comprises a valve (**14**) via which air is pumped into or released from the corresponding one of the air bags (**12**); a frame (**70**) extending around the outer perimeter of the cushion unit (**10**) such that the cushion unit substantially fills an area encompassed by the frame and comprising bores (**71**) for receiving the valves (**14**); and a superficial layer (**50**) for covering one of two opposite faces of the cushion unit (**10**).
